

WESTMINSTER
International University in Tashkent

System Analysis & Design

Tutorial 04

Agenda

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1. Previous tutorial recap
    1. Use case diagram
    2. Use case
2. Sequence diagram

Previous tutorial recap

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1. Use case diagrams
   - What does it record?
   - What shapes does it have?

2. Use cases
   - What is it?
   - In what format do we document it?

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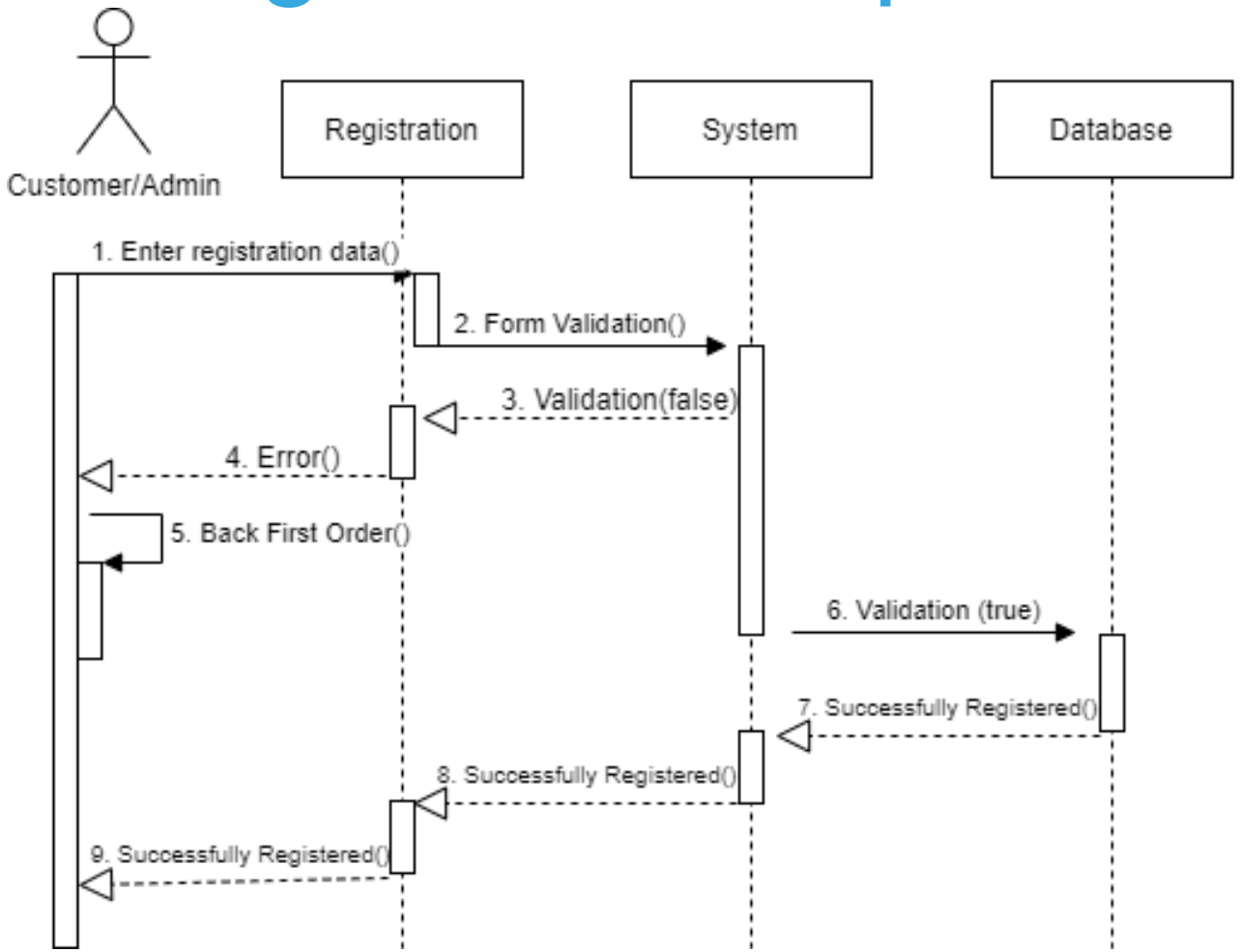
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Sequence Diagrams

Documenting object interactions in time sequence.

Sequence diagram: Example

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Nasution, M & M.Kom, Suendri & Harahap, Aninda. (2023). Customer Service Information System Using Dynamic Priority Scheduling Algorithm At PT Sumatra Sistem Integrasi. Journal of Information Systems and Technology Research. 2. 25-37. 10.55537/jistr.v2i1.324.

What is Use Case?

Google

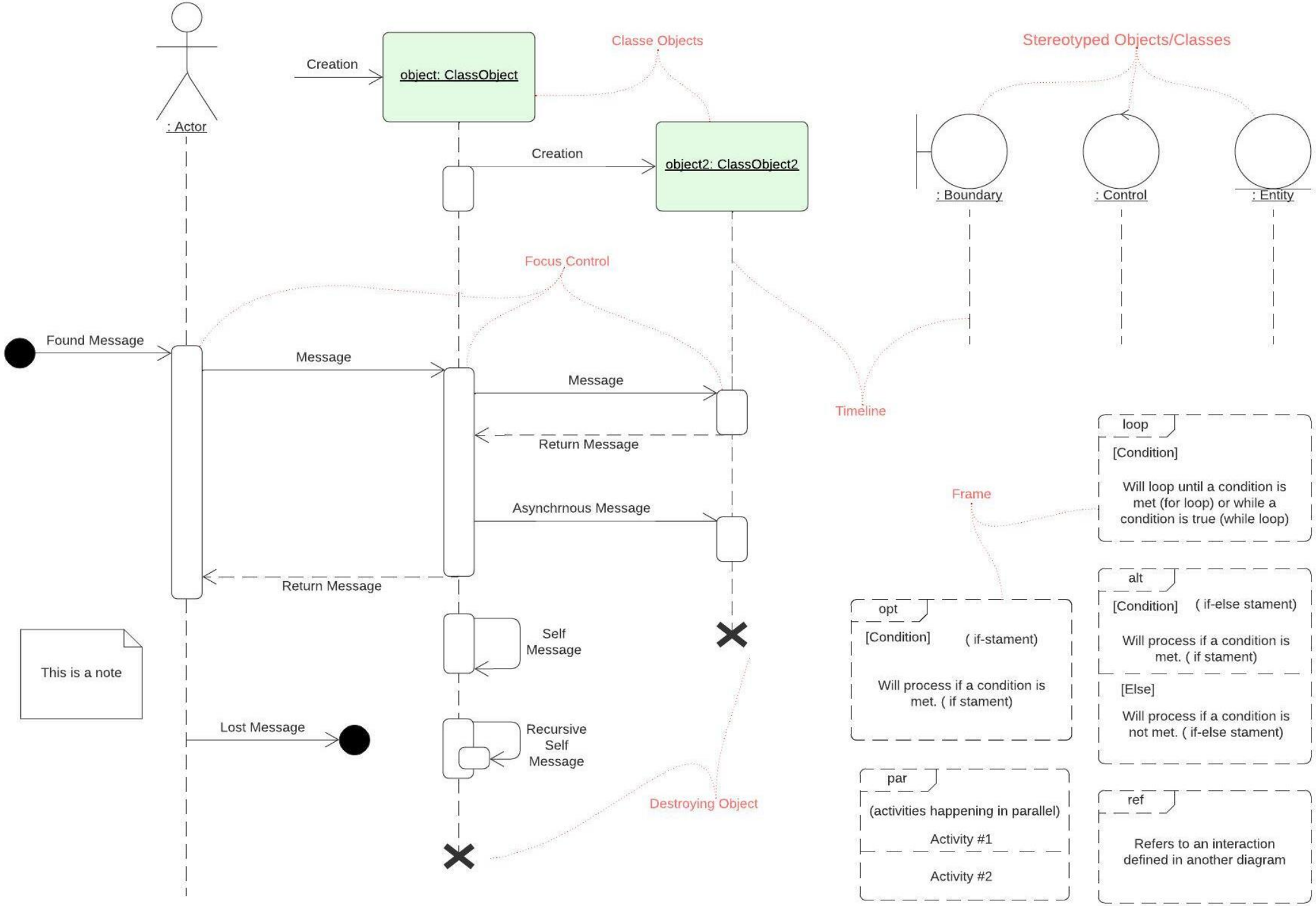
"… shows how objects or components within a system interact by illustrating the order of messages exchanged between them over time."

https://www.google.com/search?q=what+is+sequence+diagram

Wikipedia

"…shows process interactions arranged in time sequence. This diagram depicts the processes and objects involved and the sequence of messages exchanged as needed to carry out the functionality."

Source: https://en.wikipedia.org/wiki/Sequence_diagram


: Actor
Creation
object: ClassObject
Classe Objects
Creation
object2: ClassObject2
Focus Control
Found Message
Message
Message
Return Message
Asynchrnous Message
Return Message
Self
Message
This is a note
Lost Message
Recursive
Self
Message
Destroying Object
Timeline
Stereotyped Objects/Classes
: Boundary
: Control
: Entity
Frame
loop
[Condition]
Will loop until a condition is met (for loop) or while a condition is true (while loop)
opt
[Condition] (if-stament)
Will process if a condition is met. (if stament)
alt
[Condition] (if-else stament)
Will process if a condition is met. (if stament)
[Else]
Will process if a condition is not met. (if-else stament)
par
(activities happening in parallel)
Activity #1
Activity #2
ref
Refers to an interaction defined in another diagram

Sequence Diagram

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Let’s draw sequence diagram for:

- Login
- Checkout on e-commerce website
- Enrolling to modules

Source: http://www.gatherspace.com/static/use_case_example.html

Exercise: Group

1. Open “SAD-Tutorial-02-Cases.pdf” file.
2. Read the first case.
3. Create the Sequence Diagram for it on the whiteboard.

Exercise: Individual

1. Take 15 minutes to read and create a Sequence Diagram (on your own).

2. Anyone would like to draw their version on the whiteboard?

3. Discuss how your Sequence Diagram is similar or different from the one on the whiteboard.

Note: You can create it on a piece of paper or use any online tool.

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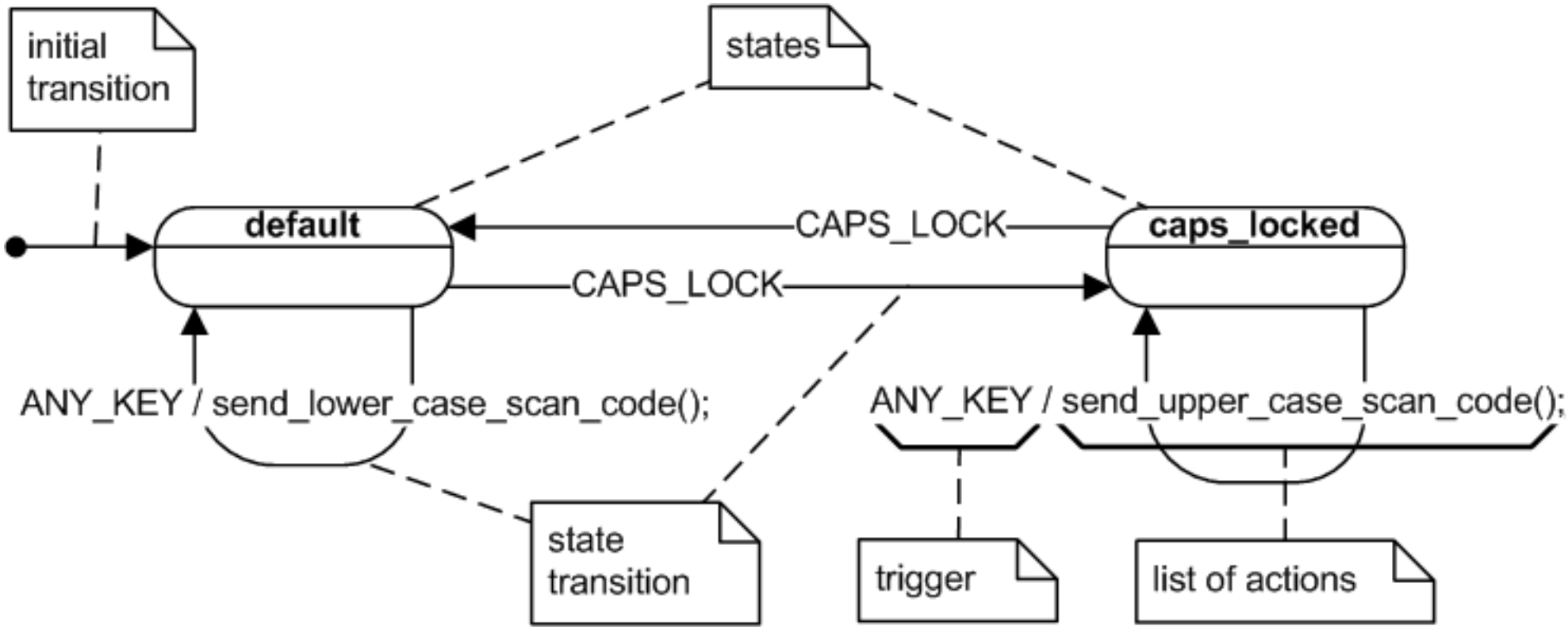
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State Machine Diagram

Documenting the states, events and transitions.

State Diagram: Example

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Source: https://en.wikipedia.org/wiki/UML_state_machine

What is State Machine Diagram?

Google

"… models the behavior of an object or system by showing its different states and the transitions between those states, which are triggered by events or signals."

https://www.google.com/search?q=state+machine+diagram

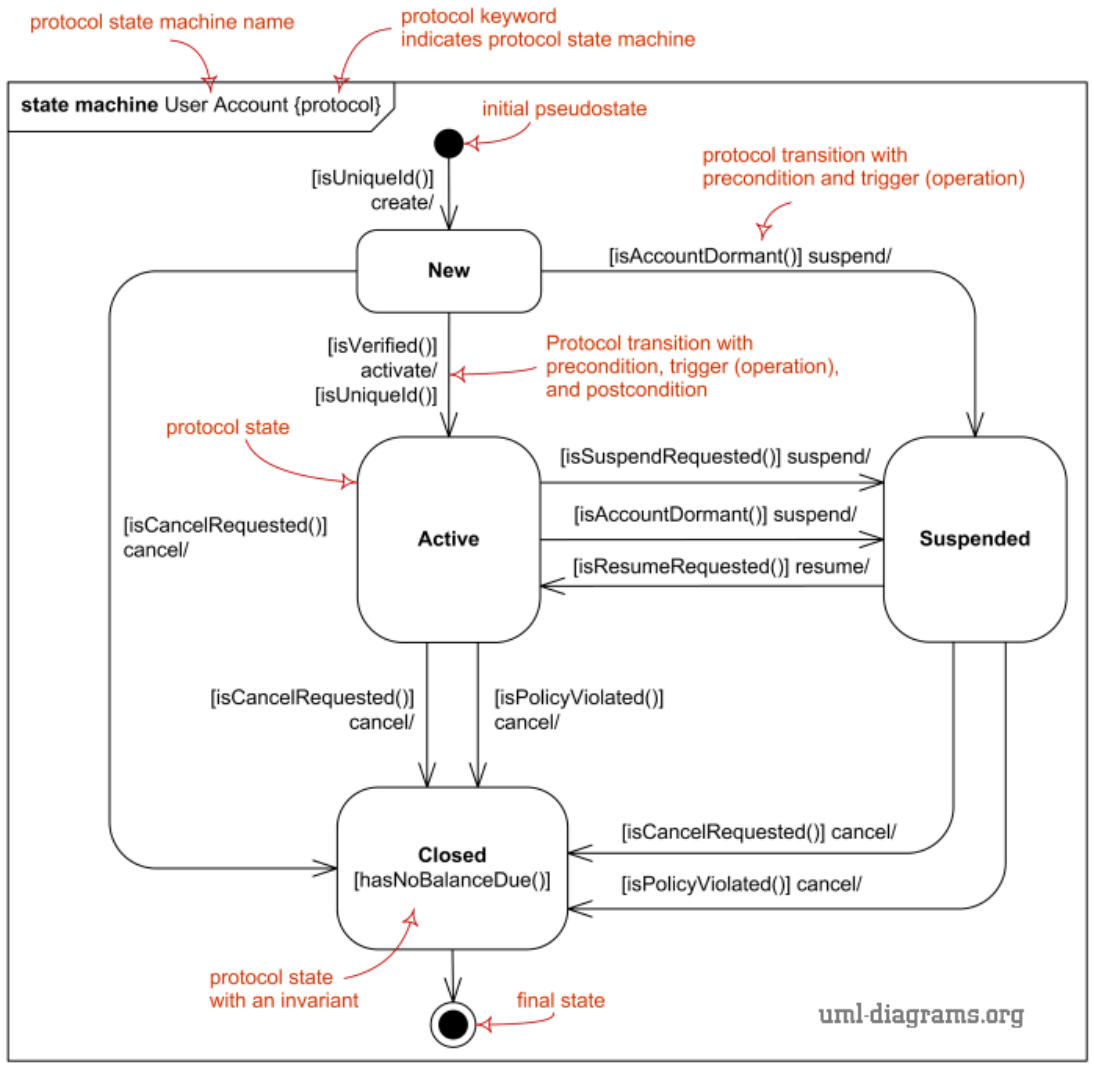
Wikipedia

"… is an extension of the mathematical concept of a finite automaton in computer science applications as expressed in the UML notation."

Source: https://en.wikipedia.org/wiki/UML_state_machine

State Machine Diagram: Shapes

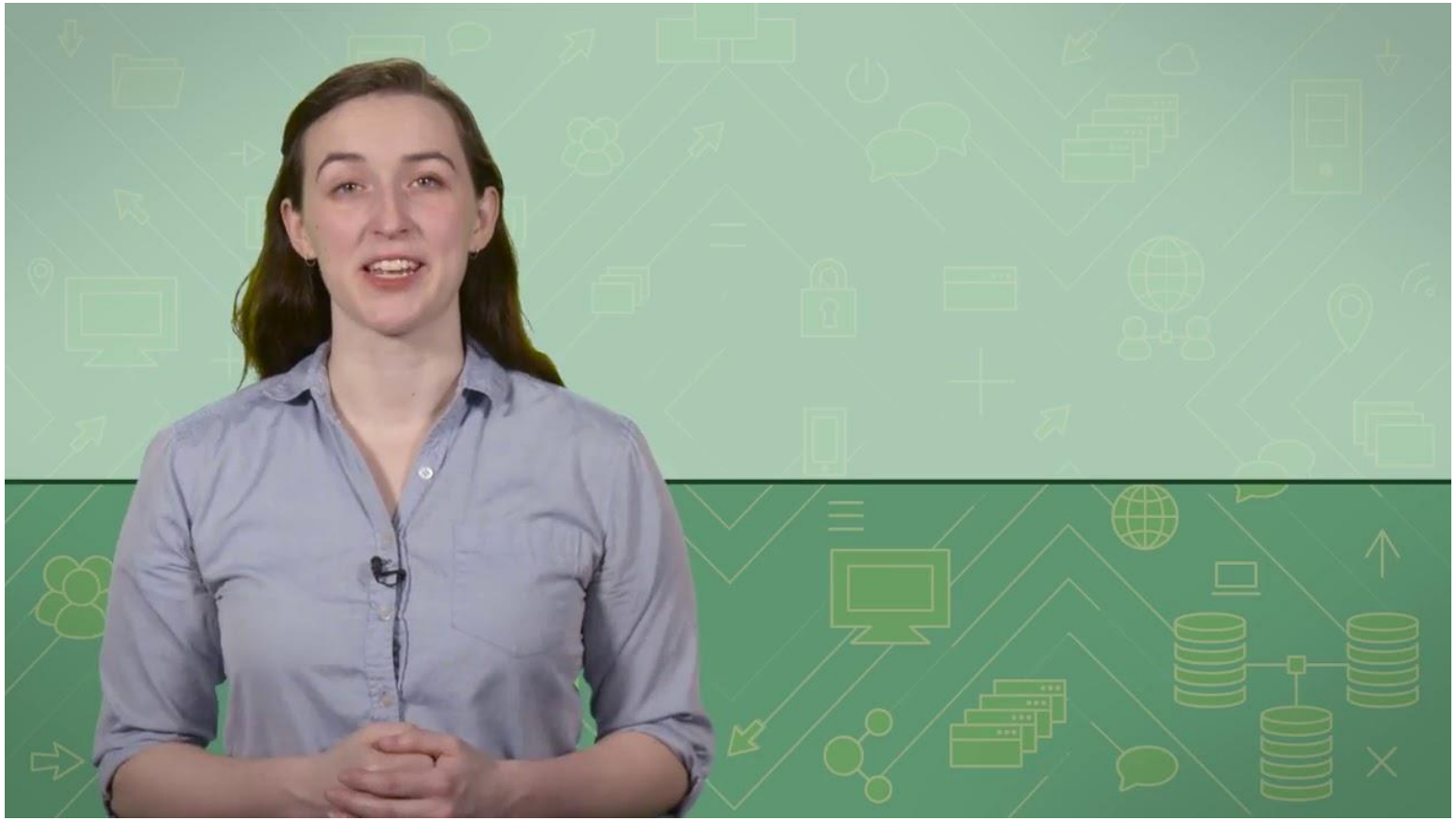
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Source: https://www.uml-diagrams.org/protocol-state-machine-diagrams.html

State Machine Diagram: Shapes

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Exercise: Group

1. Open “SAD-Tutorial-02-Cases.pdf” file.
2. Read the first case.
3. Create the State diagram for it on the whiteboard.

Exercise: Individual

1. Take 15 minutes to read the second case and draw its State Diagram (on your own).

2. Anyone would like to draw their version on the whiteboard?

3. Discuss how your State diagram is similar or different from the one on the whiteboard.

Note: You can create it on a piece of paper or use any online tool.

The End

See you at the lecture.